# Individual Progress Report: Week 3

**Group Topic:** Urban Air Quality & Environmental Health **Lab No.** Week 3

**Git Repo/ Colab Link:** [https://github.com/Cristy16/BDA---Final-Project](https://github.com/Cristy16/BDA---Final-Project) **Date:** 05/08/26

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## I. Objectives

### General Objectives

To establish a high-quality, machine-readable dataset through structured preprocessing and data splitting, ensuring the machine learning model is trained on accurate information for pollution prediction.

### Individual Objective (Setup and Justification)

- **Data Preparation:** Perform a final round of cleaning on the sampled dataset to remove any inconsistencies or missing values that could skew model results.
- **Feature Transformation:** Implement One-Hot Encoding for categorical location data to convert station names into a numerical format that the algorithm can interpret.
- **Model Integrity:** Divide the processed data into training and testing subsets (80/20 split) to ensure the model is evaluated on unseen data, preventing simple memorization.

## II. Tasks Completed

- **Final Data Verification:** I audited the 500,000-row dataset to ensure all primary features (CO, NO_2, SO_2) and the target variable (PM_2.5) were purely numerical and free of null entries.

- **Categorical Encoding:** Since machine learning models cannot process raw text like "Station A," I applied **One-Hot Encoding** to the stationname column. This created binary columns for each location, allowing the model to factor in geographical differences in air quality.

- **Dataset Partitioning:** I successfully performed an **80/20 Train-Test split**. This step is critical because it isolates 20% of the data to act as a "final exam" for the model, proving it can predict PM_2.5 levels in real-world scenarios rather than just repeating historical data.

- **Logic Implementation:** Developed the Python script using pandas and scikit-learn to automate these steps, ensuring the data remains organized and ready for the training phase.

## III. Tasks to be Completed

With the preprocessing and data splitting finalized, I will now hand over the prepared training and testing "piles" to the team members responsible for model training and hyperparameter tuning in the next phase of development.

## IV. Results and Analysis

### Model Performance Summary

| Step | Technique | Purpose |
|---|---|---|
| Data Cleaning | Dropna / Drop_duplicates | Removes "noise" and ensures the model learns from high-quality data. |
| Encoding | One-Hot Encoding | Translates station names into a format the machine can actually read. |
| Data Splitting | 80/20 Train-Test Split | Essential for a fair evaluation and to prevent the model from "cheating." |

### Analytical Insights

- **Refining Accuracy through Cleaning:** By performing a final check for missing values, I prevented potential system crashes or biased results that occur when a model encounters empty data points.
- **Location-Based Patterns:** Encoding the stationname allows the model to understand that PM_2.5 levels might be naturally higher in industrial zones compared to residential areas. This adds a "spatial" layer to our numerical gas data.
- **Generalization over Memorization:** The 80/20 split is the most important part of my task. If we used 100% of the data for training, we would have no way of knowing if the model is actually smart or just has a good memory. Keeping 20% hidden ensures we get an honest performance score.

## V. Challenges and Solutions

When I first tried to combine the encoded station names back with the gas levels, the rows didn't line up correctly because the index numbers were out of sync. I fixed this by using the .reset_index(drop=True) function in Python. This "reset" the row numbers for both the gas data and the encoded location data, allowing them to merge perfectly into one single matrix (X) without any alignment errors.

## VI. Conclusion

The preprocessing phase was a success. By cleaning the data, encoding text values into numbers, and splitting the dataset into training and testing piles, I have built a solid foundation for our machine learning pipeline. The data is now clean, structured, and ready to be used to train a reliable PM_2.5 prediction model.